

Final Project

Data Analysis

1)

Null values were dropped with dropna(), as well as the “family_member_count” variable corrected to make computations possible. The number of adults and children were split into numerical variables named “adults” and “children” respectively, then another variable was created that added these two together to get the total number of people in a household which was named “residents”. Then, a copy of the data frame was created that held only the quantitative variables, meaning that the following columns were removed from this copy: case_id, state, areaname, county, family_member_count.

2)

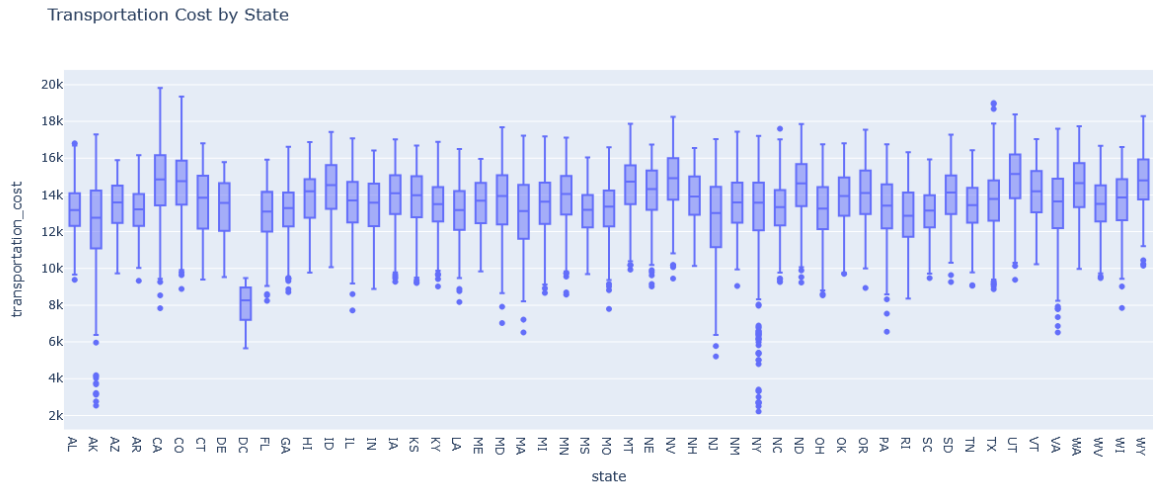
Seen here is a sample of the summary table for the dataset. It shows the data for each of the entries in a scrollable and sorted table.

case_id	state	isMetro	areaname	county	family_member_count	housing_cost	food_cost	transport	healthcare	other_needs	childcare	taxes	total_cost	median_income	adults	children	residents
1	AL	true	Montgomery AL MSA	Autauga County	1p0c	8505.72	3454.91	10829.1	5737.47	4333.81	0	6392.94	39254.0	73010.4	1	0	1
1	AL	true	Montgomery AL MSA	Autauga County	1p1c	12067.5	5091.70	11588.1	8659.55	6217.45	6147.82	7422.07	57194.3	73010.4	1	1	2
1	AL	true	Montgomery AL MSA	Autauga County	1p2c	12067.5	7460.20	12361.7	11581.6	7075.65	15824.6	9769.56	76141.0	73010.4	1	2	3
1	AL	true	Montgomery AL MSA	Autauga County	1p3c	15257.1	9952.23	13452.1	14503.7	9134.35	18802.1	13101.7	94203.5	73010.4	1	3	4
1	AL	true	Montgomery AL MSA	Autauga County	1p4c	15257.1	12182.2	13744.5	17425.7	9942.36	18802.1	13469.2	100823.	73010.4	1	4	5
1	AL	true	Montgomery AL MSA	Autauga County	2p0c	10180.7	6224.01	12861.8	11474.0	5082.78	0	8226.72	55071.6	73010.4	2	0	2

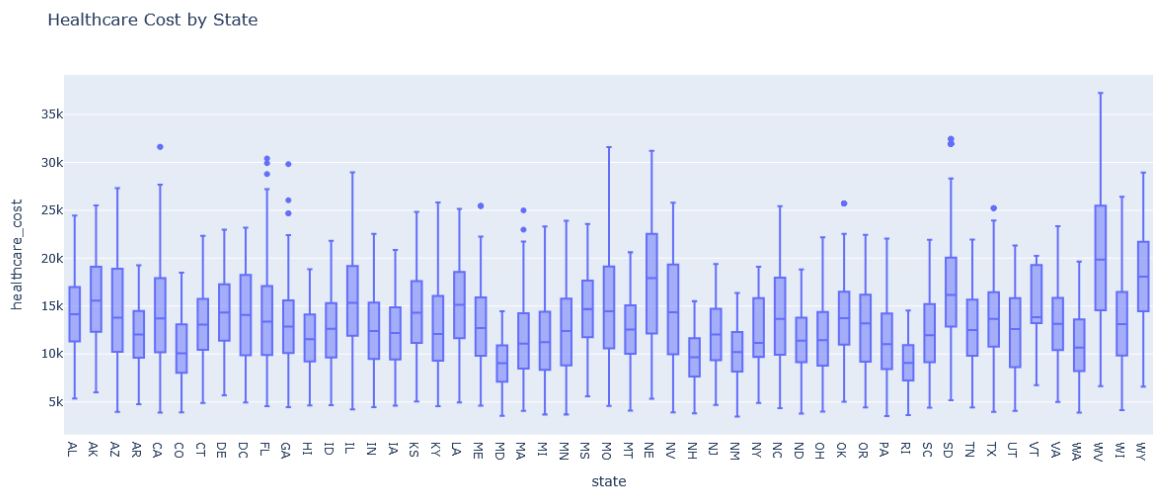
3)

- Transportation cost by state. 39 of the 51 states have outliers. Washington D.C. has a notably lower mean than the rest, though no outliers. There is some variance in

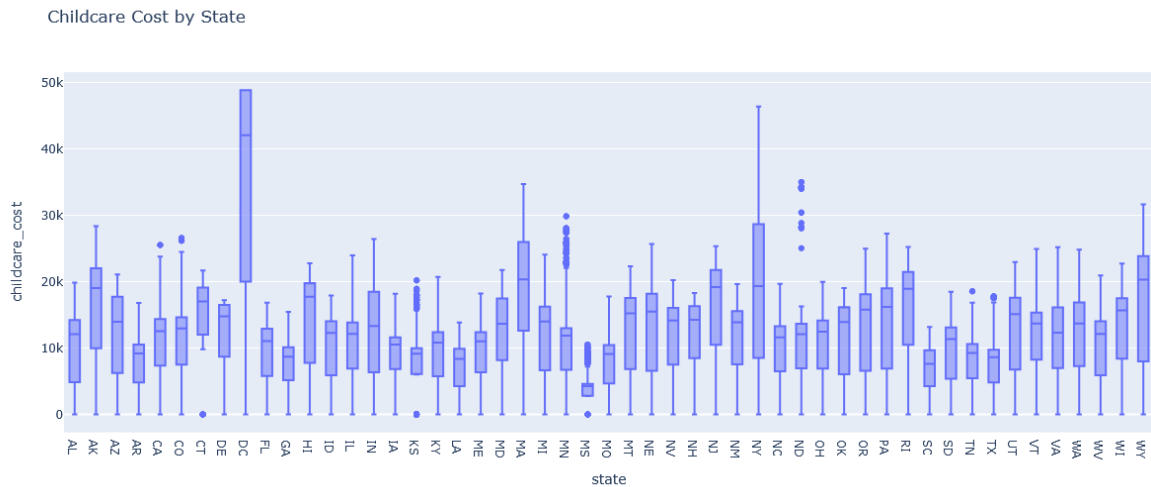
means and quartiles between states. However, most of the states have means on the higher end of the transportation_cost ranges. Interestingly, New York has many outliers that are lower than the quartiles. Out of all states, only Alaska, North Carolina, and Texas have outliers with higher transportation_cost than their quartiles. Looking at all states, Utah has the highest mean for transportation_cost.



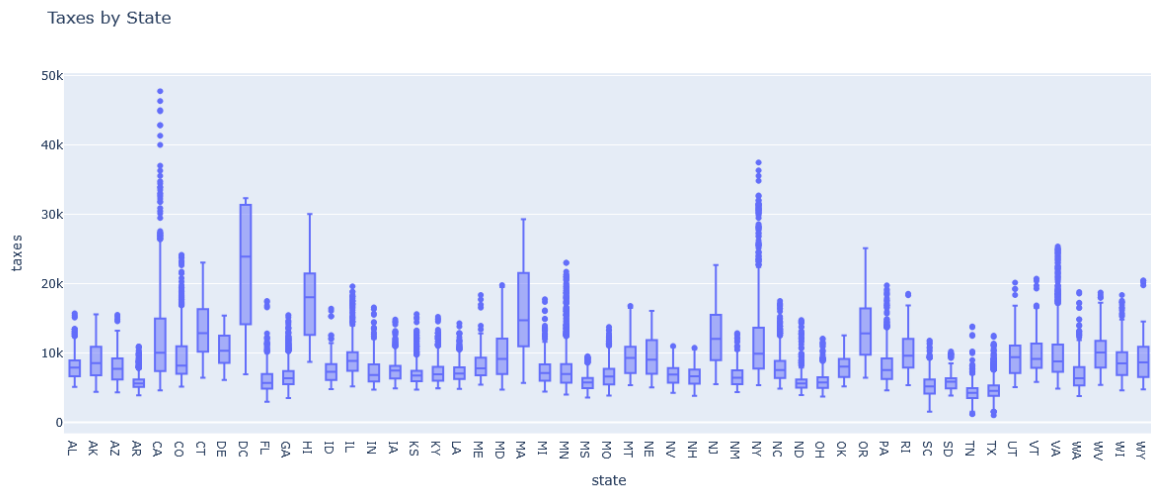
- b. Healthcare cost by state. 8 of the 51 states have outliers. The means for healthcare cost are generally similar. Georgia's quartiles are fairly low, but its outliers reach much higher. West Virginia has a notably higher mean and 4th quartile compared to the rest of the states. Maryland, New Mexico, New Hampshire, and Rhode Island all have slightly lower means than the other states.



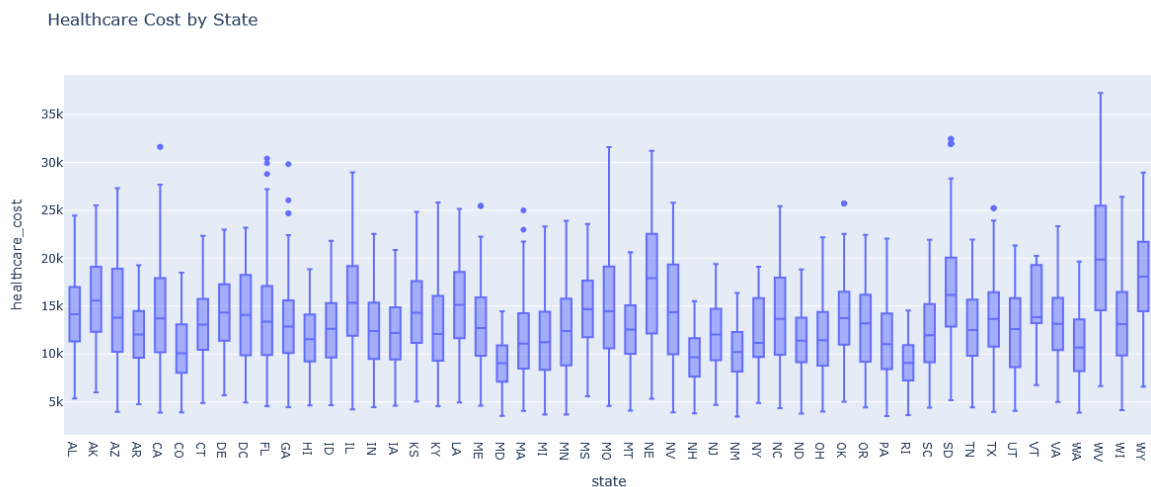
- c. Below is a box plot of childcare cost by state. 9 of the 51 states have outliers. It should be noted that the 0s for families without children were included, so some of the quartile ranges may be being influenced by that. Additionally, the 0s are the only outliers smaller than the first quartile for any state, as well as most states starting their range at 0. Washington D.C.'s mean is much higher than the rest, with its quartile ranges spanning every value for `chidcare_cost` in the dataset. On the other side, Mississippi's mean is much lower than the rest of the states, with very small quartile ranges.



- d. Below is a box plot of taxes by state. 41 of the 51 states have outliers. California's mean isn't significantly higher, though its 4th quartile range reaches high in taxes, and it has some of the highest values for taxes. Tennessee and Texas are the only states where there are outliers lower than the 1st quartile range. Arizona's quartile range is comparatively small, though it has many outliers. Washington D.C. by far has the highest mean. New York has a smaller quartile range, but also has a higher mean of taxes.

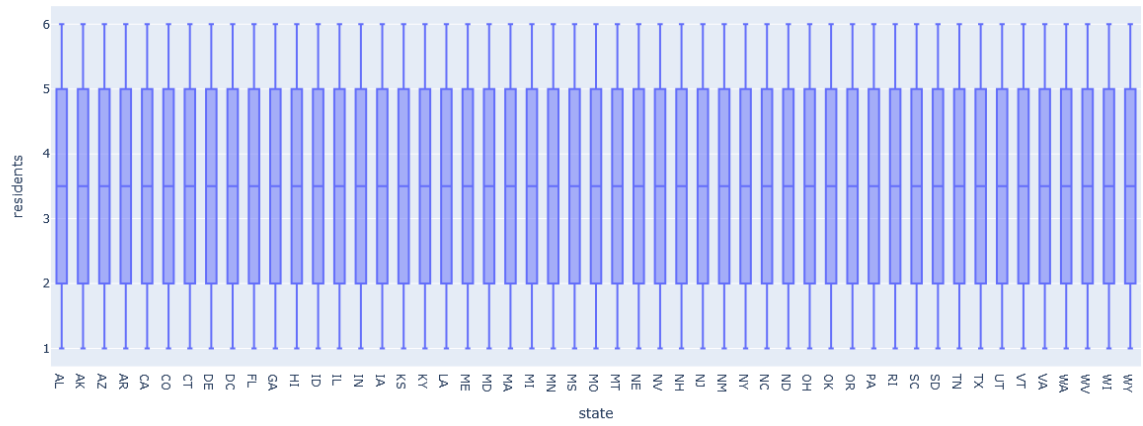


- e. Below is a box plot of healthcare cost by state. 8 of the 51 states have outliers. The means here are generally similar, not deviating too far. The quartile range for Georgia is generally lower, but it has some outliers on the higher side of healthcare_cost. West Virginia has a notably higher mean than the rest of the states. Maryland, New Mexico, New Hampshire, and Rhode Island all have means on the lower end.



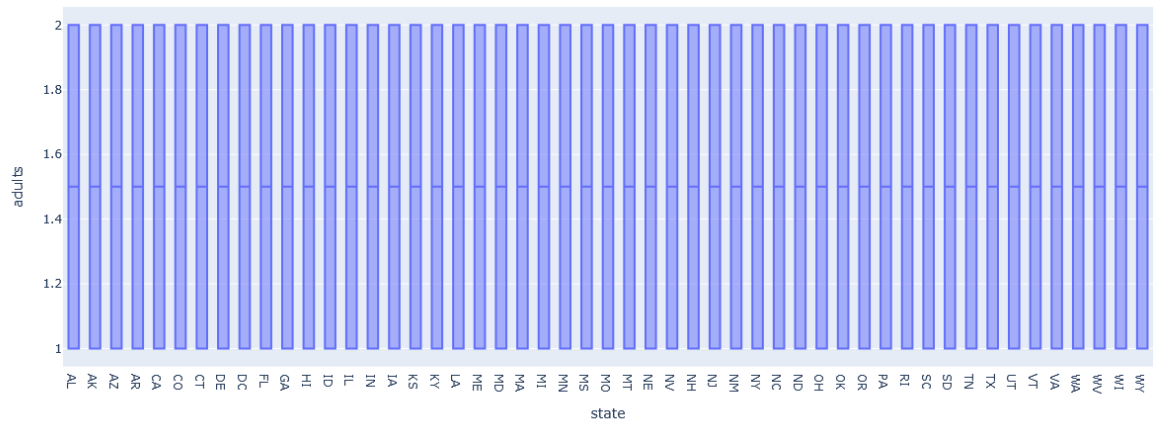
- f. Below is the box plot of residents by state. As seen, the statistics are exactly the same across all 51 states. This potentially reveals information about how the data was collected, trying to ensure the number of people interviewed was the same percentage in each state.

Total Residents by State

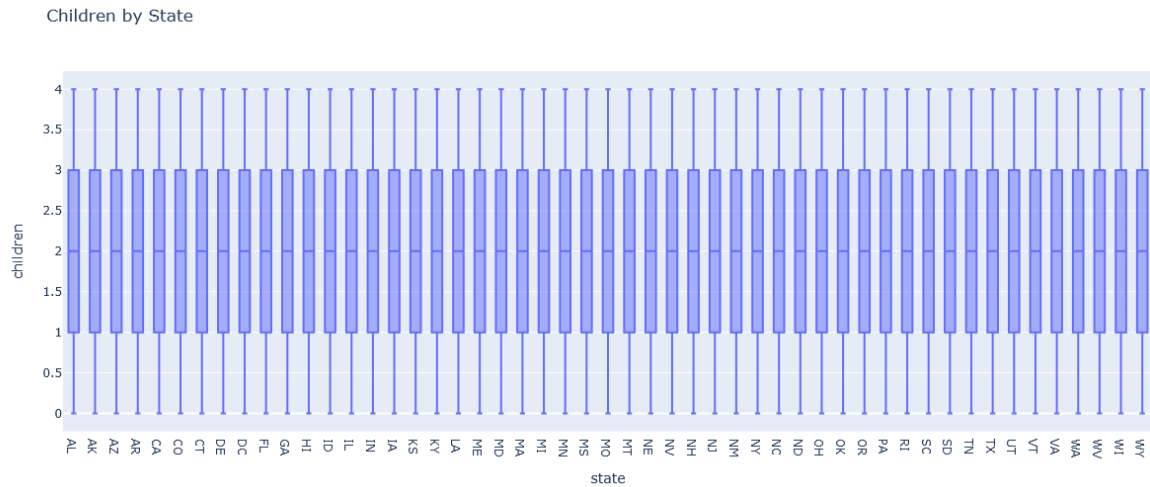


Breaking down further to adults by state, the same results are seen.

Adults by State

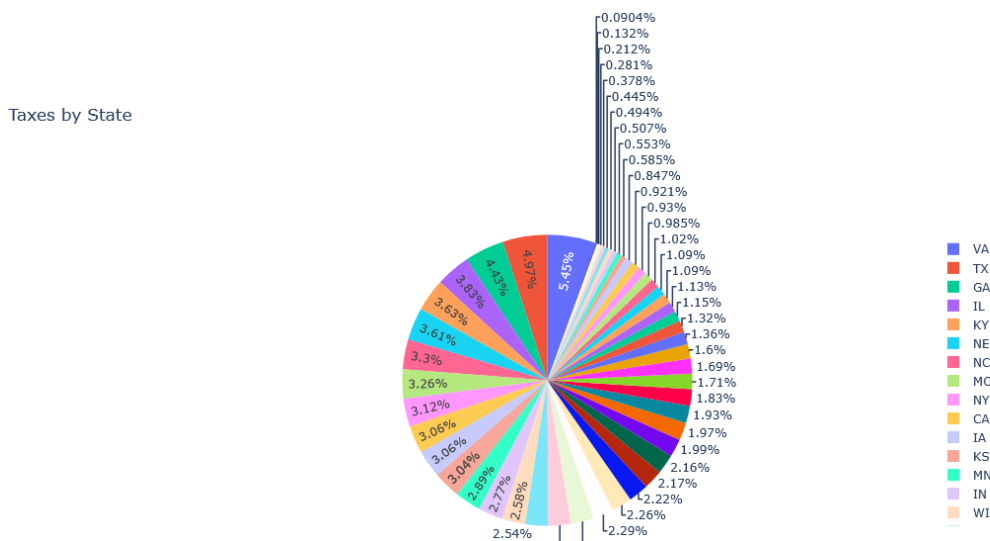


Children by state also shows the same results.



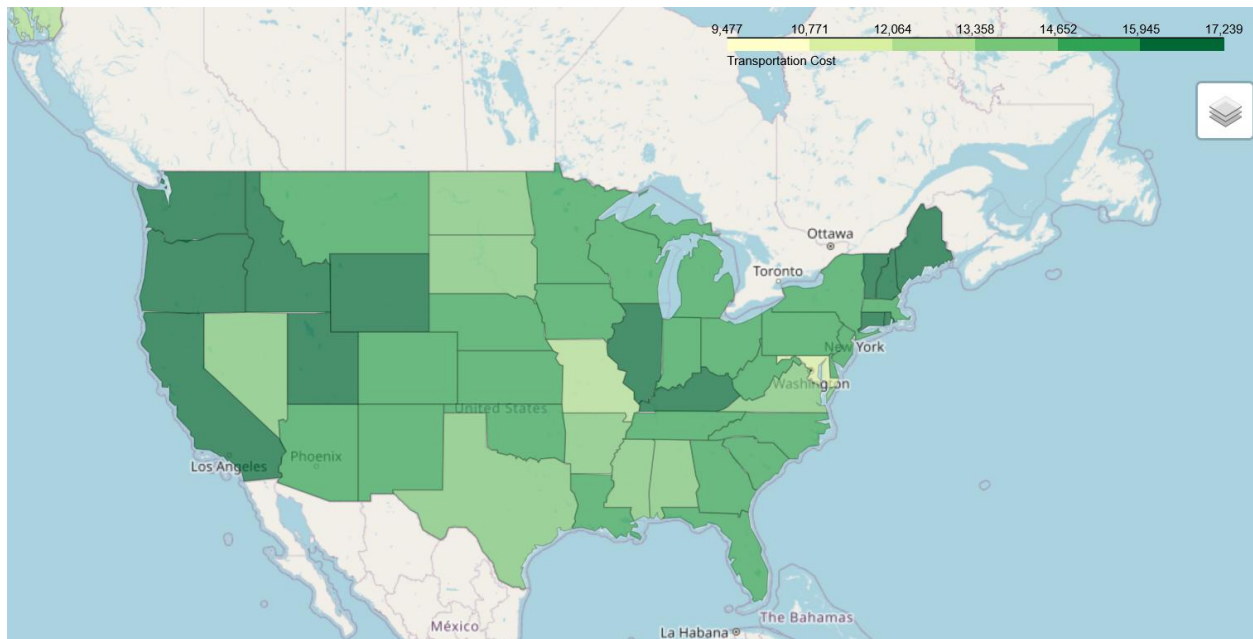
4)

Shown below is a pie chart of taxes divided by state. It can be seen Virginia makes up nearly 5.5 percent of all the taxes in this survey, with Texas and Georgia not far behind at about 5 percent and about 4.5 percent respectively. The lowest states like Washington D.C. and Delaware only make up around 0.1 percent of the total taxes surveyed. Compared to the bar chart, we see a very different story. Since this pie chart is looking at tax totals rather than averages, we see very different data. While this is useful information still, it highlights the importance of choosing the right metrics to measure, as well as the right visualization. Key disadvantages are that the state's tax totals aren't clearly shown and with 51 different options for state, the wedges can be difficult to differentiate between on the lower end.



5)

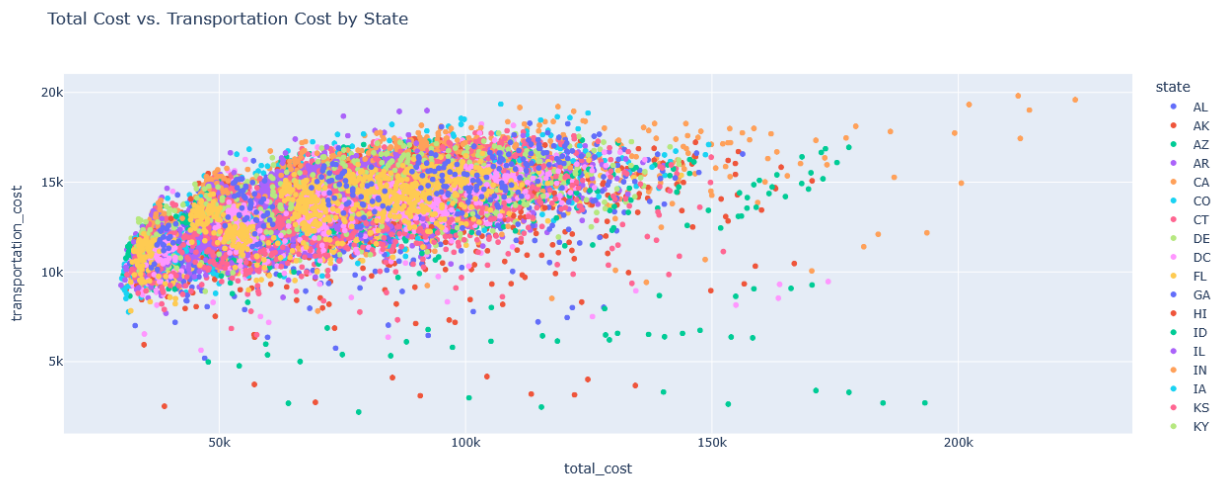
Shown below is a Choropleth map of the transportation_cost for each state. Compared to the box plot, we see similar information. It is shown again that the West Coast and New England states have higher transportation costs, while the Southern and Plains states generally have lower costs. The coloring and mapping of the Choropleth helps to more clearly show the differences between the states. However, this also suggests that the differences are much larger than the raw numbers are, which isn't an issue with the box plot.



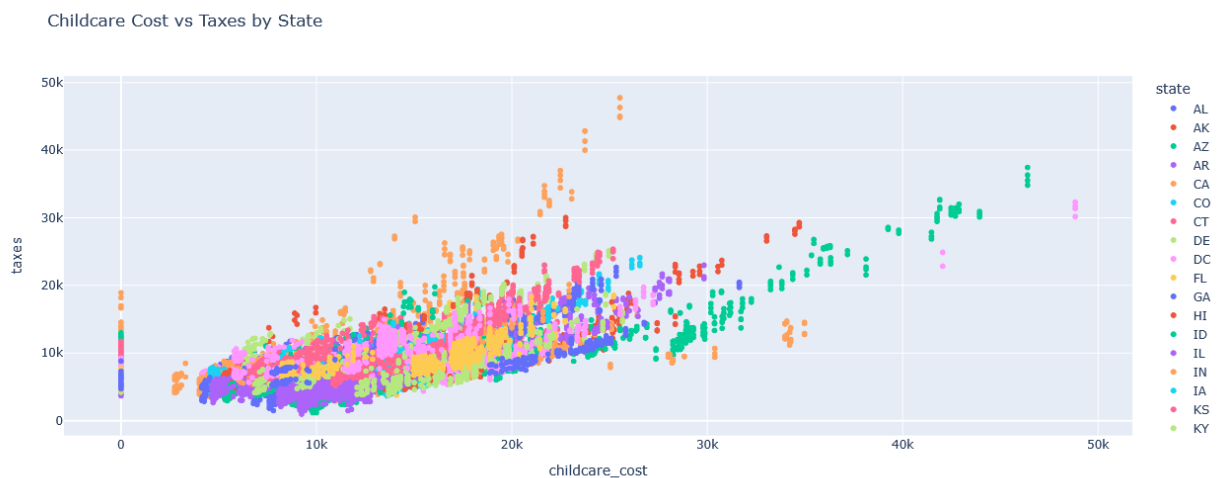
6)

- a. Shown below is a scatter plot of total_cost and transportation_cost colored by state. The correlation value between the two variables was only moderate, and this graph seems to convey that. There seems to be a possible linear trend, but the majority of the points are clustered between a little under 10000 and a little over 15000 on the y-axis, drawing the trend away from (0,0). There are a few ranges where few points lie. Very few points fall beneath 5000 for transportation_cost, and these points that do are all in Alaska and New York. Interestingly, New York also has some points on the higher end of transportation_cost, with all of its points being spread throughout. None of the points on the y-Axis reach above 20000, but the ones that get closest

are largely from California, with one from Colorado. On the x-axis, points start to get sparse after 150000, with only points from California after 200000



- b. Shown below is a scatter plot of childcare_cost and taxes colored by state. There is a string of 0s with no children that vary by state, likely skewing the graph and correlation. The points seem to form a positive linear trend. Almost all of the points that deviate upward with moderate childcare_cost but high taxes are in California. New York's trend has a similar positive linear pattern but higher childcare_cost. Some states like Texas have large clusters of points with low taxes and low childcare_cost.



- c. Shown below is a scatter plot of median_family_income and taxes colored by state. These points seem to show a moderate positive linear trend. Very few points have above 120k median_family_income and/or above 20k taxes. New York has a trend of

points that have higher taxes than other states at the same median_family_income. California has all of the highest points in taxes, which also have high median_family_income. New Mexico has a collection of points that are the highest in median_family_income but have moderate to low taxes.



- d. Shown below is a scatter plot of housing_cost and taxes colored by state. This plot seems to show a strong positive linear trend. The points on the right of the plot with both high housing_cost and high taxes all belong to California. The points for Washington D.C. have a trend of having higher taxes for other points with the same housing_cost. New York also follows a trend of having higher taxes than points with the same housing_cost, while Texas tends to have lower taxes than points with the same housing_cost.



7)

Shown below is the top 12 counties for taxes. Washington D.C. appears as 11 on this list. All of the top 10 are in either California or New York. These suggest that taxes are generally higher in these two states.

county	taxes
San Mateo County	34738.964880
Marin County	31637.529360
San Francisco County	27174.135960
New York County	26517.054360
Santa Clara County	24651.857760
Rockland County	23463.096600
Queens County	22807.598208
Norfolk County	22347.509448
Westchester County	22292.750676
Suffolk County	22256.451666
District of Columbia	21744.785220
Honolulu County	21423.277428

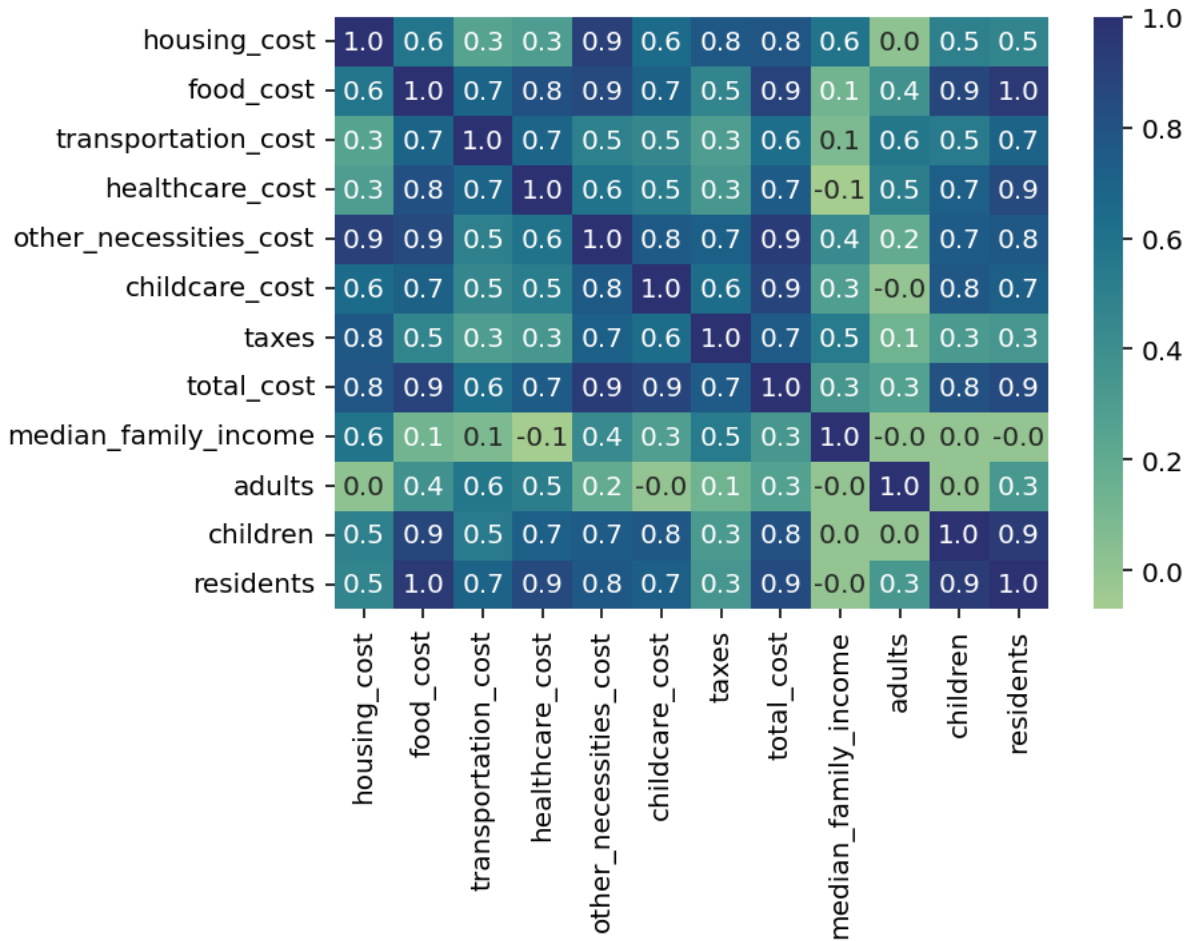
8)

Shown below are the top 12 counties for childcare_cost. Washington D.C. comes out as well and above the highest childcare cost. Other than that, the rest of these counties are in New York. This seems to suggest that New York generally has much higher healthcare cost than any other state.

county	childcare_cost
District of Columbia	31940.492400
New York County	29620.246800
Queens County	28050.334320
Rockland County	27260.377200
Westchester County	26661.030720
Suffolk County	25288.633680
Bronx County	24340.542960
Norfolk County	23606.173680
Dutchess County	22844.859600
Ulster County	21947.080944
Tompkins County	21589.945152
Grand Forks County	21525.153984

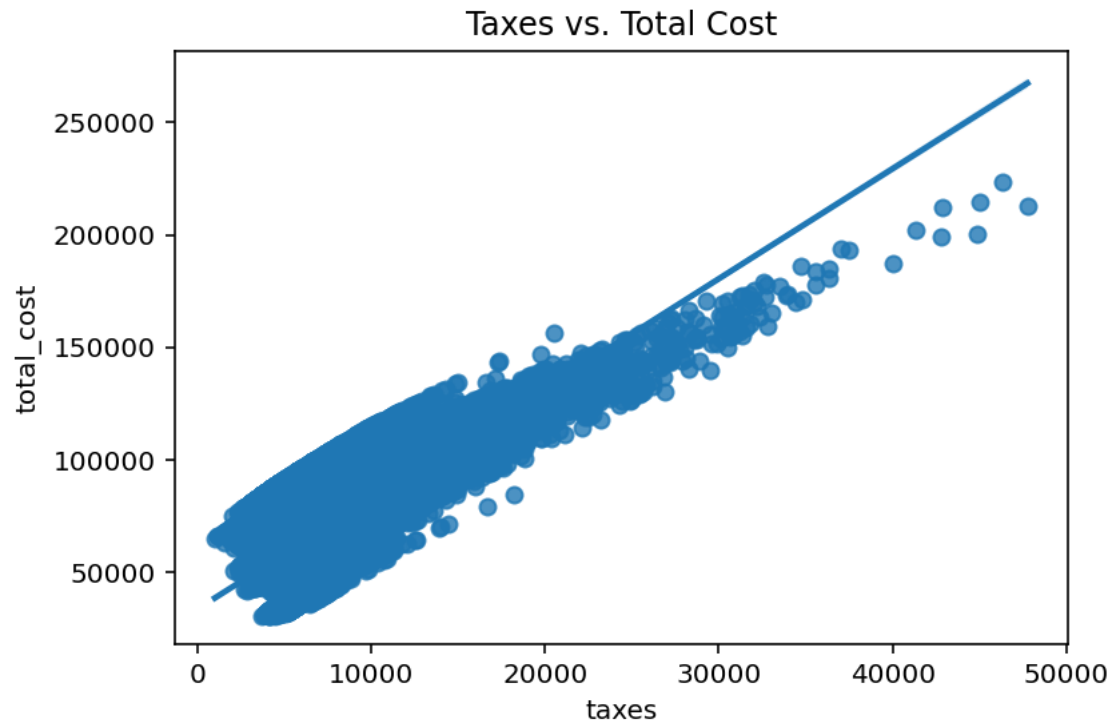
9)

The correlation heatmap for the quantitative variables in the dataset is a valuable tool for seeing correlation between any two of these variables. Key findings from this matrix are the high and low correlation values. food_cost has a full correlation of 1.0 with residents, as well as a correlation of 0.9 with total_cost, children, and other_necessities_cost. housing_cost also has a 0.9 correlation with other_necessities_cost and also has a 0.8 correlation with taxes and total_cost. There are many very low correlations, with some even being 0. To call a few notable examples, median_family_income having a very slight negative correlation with healthcare_cost, which is the only proper negative correlation. Additionally, median_family_income has a low correlation with total_cost and many of the other costs, as well as having no correlation with any of the resident variables. This heatmap does also contain some oddities. adults only has a 0.3 correlation with residents despite being part of the residents variable. This can also be seen to a lesser extent with total_cost having only a 0.6 correlation with transportation_cost, despite the latter being part of the former.



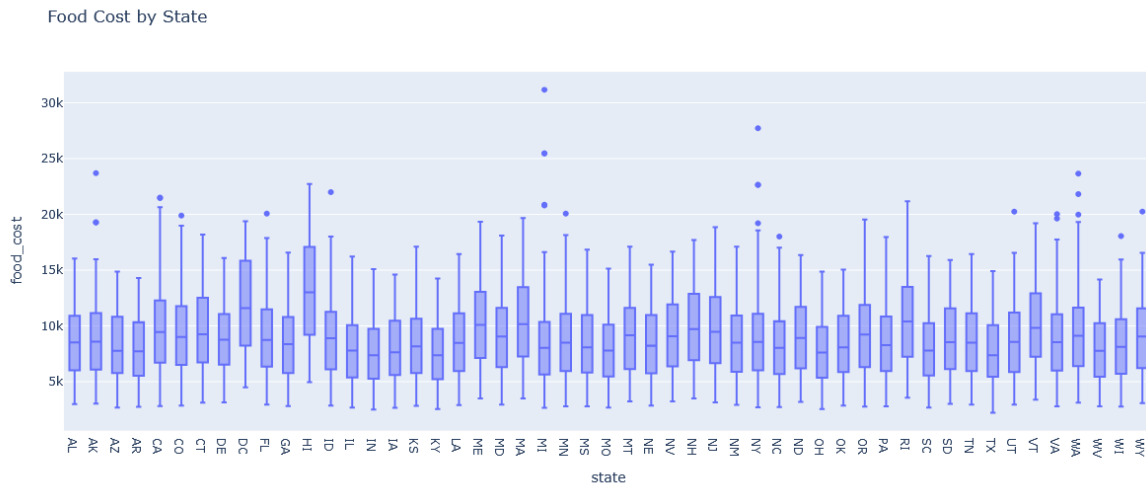
10)

The Taxes vs. Total Cost regression plot examines the potential relationship between those two variables. This graph shows a strong cluster in the 0-20000 range on taxes and 0-100000 for total_cost. The regression line also seems to clearly follow a positive linear relationship for the first half, but then the points tail off below the line at the end. This is likely influenced by the large cluster at the beginning.



11)

Shown below is a box plot of food_cost by state. Something that can be seen immediately is that there are 22 points that are potential outliers spread across 14 states. These outliers are all on the higher end of food_cost. Interestingly, the single highest food_cost point which is above 30000, is in Michigan, which has fairly low mean food_cost. Michigan is also tied with New York and Washington for the highest amount of outliers, being 3. Michigan's, however, are all far above the upper bound of the quartiles.



12)

Shown below is a bubble scatter plot of food_cost and taxes sized by childcare_cost. One thing that immediately stands out is that there is a strong trend of the non-metro states being on the lower end of taxes and the metro states being on the higher end of taxes. There are some points that are non-metro that are a bit higher in taxes, but all of the highest taxes points are metro. The size of childcare_cost does seem to follow a trend of getting larger as the food_cost and taxes increase, at least at the beginning of the plot. There are also a moderate positive linear trend that could be seen. The vast majority of points are between 5000 and 20000 food_cost and 0 and 30000 taxes.



13)

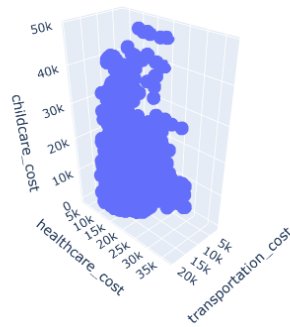
Shown below is a bubble scatter plot of housing_cost and taxes sized by transportation cost and colored by isMetro. One thing immediately seen is that metro cases have points all across the graph, but non-metro states are limited to the lower half of the range of both housing_cost and taxes. There is also a seemingly strong linear trend between housing_cost and taxes. There also appears to be no strong correlation between transportation_cost and the other variables.



14)

Shown below is a 3D scatter plot of transportation_cost, healthcare_cost, and childcare_cost. One thing that can immediately be seen is that the vast majority of the points are on the higher end of transportation_cost. This makes analysis of the plot difficult, with a low spread of points. Because of this, many strong outcomes can't be gleaned from it. Despite this, one thing we can see is that many of the points that have 0 childcare_cost have high healthcare_cost and a high variance in transportation_cost.

Transportation Cost vs. Healthcare Cost vs. Childcare Cost



15)

Shown below is a 3D scatter plot of childcare_cost, median_family_income, and taxes, colored by state. One notable standout is that there is a cluster of points from Virginia that have low childcare_cost, high median_family_income, and high taxes. The points with 0 childcare_cost all have varied median_family_income and low taxes.

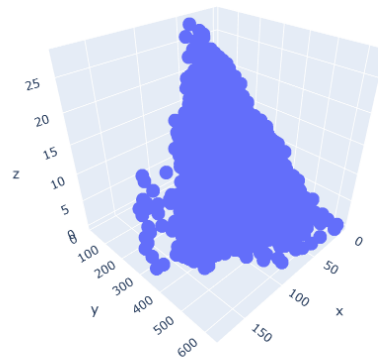
Childcare Cost vs. Taxes vs. Median Family Income by State



16)

Shown below is a 3D scatter plot of an NMF for this dataset using 3 components. There is a general trend seen here of component 3 being higher when component 1 is lower. There is also a trend of component 2 being higher when component 3 is lower.

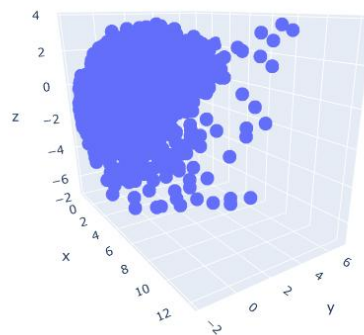
3 Component NMF



17)

Shown below is a 3D scatter plot for a 3 component PCA of the dataset. These points have a large spread, not leading to very much conclusive information.

3 Component PCA



Conclusion

Looking at this data, there are a few takeaways that can be made. One major one is the prevalence of Washington D.C. having different means than the other states, either higher or lower. On the topic of states, there is a trend of the costs being higher in the West Coast and New England states, while the Southern and Plains states generally have lower costs. Costs themselves generally are correlated notably, especially total_cost with all of the

other cost-related variables. Metropolitan areas seem to have generally higher taxes than areas that are non-metropolitan.

Appendix

```
# Schlomann_Final02
```

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
import plotly.express as px
```

```
import plotly.graph_objects as go
```

```
import folium
```

```
import requests
```

```
from sklearn.preprocessing import StandardScaler
```

```
from sklearn.decomposition import PCA
```

```
from sklearn.decomposition import NMF
```

```
df = pd.read_csv("C:\\PATH\\TO\\cost_of_living_us.csv")
```

```
# 1)
```

```
df = df.dropna()
```

```
# Gemini helped
```

```
df["adults"] = (df["family_member_count"].str[0]).astype(int)
```

```
df["children"] = (df["family_member_count"].str[2]).astype(int)
```

```
df["residents"] = (df["adults"]+df["children"])
```

```
df_quant = df.drop(columns = ["case_id", "isMetro", "state", "areaname", "county",  
"family_member_count"])
```

```

df_sample = df.sample(frac=0.1, random_state=1)

df_sample.to_csv("C:\\PATH\\TO\\cost_of_living_us_sample.csv")

# 2)

fig2 = go.Figure(data=[go.Table(
    header=dict(values=list(df.columns),
        fill_color='paleturquoise',
        align='left'),
    cells=dict(values=[df.case_id, df.state, df.isMetro, df.arenaname, df.county,
        df.family_member_count, df.housing_cost, df.food_cost,
        df.transportation_cost, df.healthcare_cost, df.other_necessities_cost,
        df.childcare_cost, df.taxes, df.total_cost,
        df.median_family_income, df.adults, df.children, df.residents],
        fill_color='lavender',
        align='left'))
])

fig2.write_html("C:\\PATH\\TO\\Final_Summary.html")

# 3)

fig31 = px.box(df, x="state", y="transportation_cost", title="Transportation Cost by State")
fig31.write_html("C:\\PATH\\TO\\Final_Boxplot31.html")

fig32 = px.box(df, x="state", y="healthcare_cost", title="Healthcare Cost by State")
fig32.write_html("C:\\PATH\\TO\\Final_Boxplot32.html")

fig33 = px.box(df, x="state", y="childcare_cost", title="Childcare Cost by State")
fig33.write_html("C:\\PATH\\TO\\Final_Boxplot33.html")

```

```
fig34 = px.box(df, x="state", y="taxes", title="Taxes by State")
```

```
fig34.write_html("C:\\PATH\\TO\\Final_Boxplot34.html")
```

```
fig35 = px.box(df, x="state", y="median_family_income", title="Median Family Income by State")
```

```
fig35.write_html("C:\\PATH\\TO\\Final_Boxplot35.html")
```

```
fig36 = px.box(df, x="state", y="residents", title="Total Residents by State")
```

```
fig36.write_html("C:\\PATH\\TO\\Final_Boxplot36.html")
```

```
fig36a = px.box(df, x="state", y="adults", title="Adults by State")
```

```
fig36a.write_html("C:\\PATH\\TO\\Final_Boxplot36a.html")
```

```
fig36b = px.box(df, x="state", y="children", title="Children by State")
```

```
fig36b.write_html("C:\\PATH\\TO\\Final_Boxplot36b.html")
```

```
# 4)
```

```
fig4 = px.pie(df, values='taxes', names="state", title="Taxes by State")
```

```
fig4.update_layout(margin=dict(t=0, b=0, l=0, r=0))
```

```
fig4.write_html("C:\\PATH\\TO\\Final_Pie4.html")
```

```
# 5)
```

```
state_geo_url = "https://raw.githubusercontent.com/python-visualization/folium-example-data/main/us_states.json"
```

```
state_geo = requests.get(state_geo_url).json()
```

```
m = folium.Map(location=[48, -102], zoom_start=4)
```

```
folium.Choropleth(  
    geo_data=state_geo,  
    name="Transportation Cost by State",  
    data=df,  
    columns=['state', 'transportation_cost'],  
    key_on="feature.id",  
    fill_color="YlGn",  
    fill_opacity=0.7,  
    line_opacity=0.2,  
    legend_name="Transportation Cost",  
)
```

```
folium.LayerControl().add_to(m)
```

```
m.save("C:\\PATH\\TO\\Final_Choropleth5.html")
```

```
# 6)
```

```
fig61 = px.scatter(df, x="total_cost", y="transportation_cost",  
                  color="state", title="Total Cost vs. Transportation Cost by State")
```

```
fig61.write_html("C:\\PATH\\TO\\Final_Scatter61.html")
```

```
fig62 = px.scatter(df, x="childcare_cost", y="taxes",  
                  color="state", title="Childcare Cost vs Taxes by State")
```

```
fig62.write_html("C:\\PATH\\TO\\Final_Scatter62.html")
```

```
fig63 = px.scatter(df, x="median_family_income", y="taxes",  
                  color="state", title="Median Family Income vs. Taxes by State")
```

```
fig63.write_html("C:\\PATH\\TO\\Final_Scatter63.html")
```

```
fig64 = px.scatter(df, x="housing_cost", y="taxes",  
                  color="state", title="Housing Cost vs. Taxes by State")
```

```
fig64.write_html("C:\\PATH\\TO\\Final_Scatter64.html")
```

```
# 7)
```

```
df_childcare = df.loc[:, ['childcare_cost', 'county']]
```

```
childcare_county = df_childcare.groupby(['county']).mean()
```

```
childcare_county.sort_values(by='childcare_cost', ascending=False).head(12)
```

```
# 8)
```

```
df_taxes = df.loc[:, ['taxes', 'county']]
```

```
taxes_county = df_taxes.groupby(['county']).mean()
```

```
taxes_county.sort_values(by='taxes', ascending=False).head(12)
```

```
# 9)
```

```
correlation = df_quant.corr()
```

```
sns.heatmap(df_quant.corr(), annot=True, fmt=".1f", cmap="crest")
```

```
# 10)
```

```
sns.regplot(x="taxes", y="total_cost", data=df)
plt.title("Taxes vs. Total Cost")
plt.show()
```

```
# 11)
```

```
fig11 = px.box(df, x='state', y='food_cost', title="Food Cost by State", hover_data='case_id')
fig11.write_html("C:\\PATH\\TO\\Final_Boxplot11.html")
```

```
# 12)
```

```
fig12 = px.scatter(df, x="food_cost", y="taxes",
                    size="childcare_cost", color="isMetro", title="Food Cost vs. Taxes sized by
Childcare Cost")
fig12.write_html("C:\\PATH\\TO\\Final_BubbleChart12.html")
```

```
# 13)
```

```
fig13 = px.scatter(df, x="housing_cost", y="taxes",
                    size="transportation_cost", color="isMetro", title="Housing Cost vs. Taxes sized
by Transportation Cost")
fig13.write_html("C:\\PATH\\TO\\Final_BubbleChart13.html")
```

```
# 14)
```

```
fig14 = px.scatter_3d(df, x='transportation_cost', y='healthcare_cost',
                       z='childcare_cost', title="Transportation Cost vs. Healthcare Cost vs. Childcare
Cost")
fig14.write_html("C:\\PATH\\TO\\Final_Scatter3d14.html")
```

```
# 15)
```



```
fig15 = px.scatter_3d(df, x='childcare_cost', y='taxes',  
                      z='median_family_income', color='state', title="Childcare Cost vs. Taxes vs.  
Median Family Income by State")
```

```
fig15.write_html("C:\\PATH\\TO\\Final_Scatter3d15.html")
```

```
# 16)
```

```
scaler = StandardScaler()
```

```
df_scaled = scaler.fit_transform(df_quant)
```

```
pca = PCA(n_components=3)
```

```
df_pca = pca.fit_transform(df_scaled)
```

```
plt.figure()
```

```
fig16 = px.scatter_3d(x=df_scaled[:, 0], y=df_scaled[:, 1], z=df_scaled[:, 2], title="3  
Component PCA")
```

```
fig16.write_html("C:\\PATH\\TO\\Final_pca.html")
```

```
# 17)
```

```
model = NMF(n_components=3, init='random', random_state=0)
```

```
W = model.fit_transform(df_quant)
```

```
H = model.components_
```

```
fig17 = px.scatter_3d(x=W[:, 0], y=W[:, 1], z=W[:, 2], title="3 Component NMF")
```

```
fig17.write_html("C:\\PATH\\TO\\Final_nmf.html")
```